

Response Letter with Annotations

Citation (APA format, using Zotero or other bibliography software)

Jaidka, K., Zhou, A., & Lelkes, Y. (2019). Brevity is the soul of Twitter: The constraint affordance and political discussion. *Journal of Communication*, 69(4), 345-372.

Annotations

1. Purposes

a. Motivation (one-sentence summary)

The motivation of this study is to explore whether change of technological affordances, constraints on message length, in Twitter, will impact the online political discussion quality

b. Specific Rationale (several bullet points)

- Previous studies held political discussions on Twitter replete with low quality discourses, such as toxic and abusive responses. They are not ideal for political deliberation.
- Technological affordances on social media platforms can influence the quality of political discussions on social media.
- Previous studies on how one particular technological affordance, message length, impacts political discussion quality presented conflicting findings.
- Twitter changed the constraints on message length from 140 characters to 280 characters per tweet.
- Civility, interactivity, and argumentation were identified as key components of high quality political deliberation.

c. Research Questions and/or Hypotheses

- RQ1: What effect does doubling the character limit have on incivility
- H1: Comments posted on Twitter after the 280-character limit change will be less informal(more polite) than before the change.
- H2: Comments posted on Twitter after the 280-character limit change will comprise a better quality of (a) arguments and (b) interactions than before the change.
 - Identify IV, DV, mediator, moderator
- IV: Twitter message length (Comments posted before or after the 280-character limit change)
- DV: incivility, formality, arguments(deliberative quality?), interactions
 - Identify specific relationships among variables

2. Method

a. Research Design

- Experiment: conditions, manipulations, etc.
- Survey: cross-sectional, longitudinal, etc.
- Content analysis: data source, etc.
- ...

The study used an interrupted times series design and an automated content analysis of over 350,000 tweets related to politics.

- Dataset:
 - Political tweets were identified as replies to 536 U.S. Congressmen and Congresswomen who were in the office before 7 November 2017.
 - Filtered and collected a Twitter 1% sample using API. Collected all the political tweets between 1 January 2017 and 31 March 2018.
 - Data was cleaned by removing retweets and tweets by 1% Twitter users subjected to early access of character limit change
 - The final dataset comprised 358.242 replies
 - Categorized tweets as compliers (users who tweeted more than 145 characters at least once after the character limit change) and the non-compliers (users who never tweeted more than 145 characters)
- Automated content analysis
 - Use self-developed supervised machine learning models, or models or dictionaries developed by other scholars to label the data
- Interrupted time series regression model

b. Sample (representative sample?)

c. Operationalization and Measures

- Politics-related or not: a supervised machine-learning classifier, trained on 6000 hand-annotated tweets; Label tweets as relevant to politics (1) or not (0).
- Incivility:
 - daily number of uncivil tweets and the average percentage proportion of uncivil words
 - Four models of incivility were applied and compared
 - **Incivility** (developed by this study): A supervised machine-learning classifier trained on 6000 hand-annotated tweets; labeled as uncivil or civil (1/0)
 - Offensive classifier: trained by dataset by Davidson et al. (2017) and labeled tweets as offensive(1) or inoffensive(0)
 - Uncivil words dictionary by Muddiman et al. (2018): the proportion of offensive words in a tweet (0 to 100 continuous values)
 - Swear words dictionary from LIWC: the proportion of swear words in a tweet (0 to 100 continuous values)
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- Formality:
 - The decrease in the use of **informal words**
 - Informal words are operationalized as the proportionate use of informal language, such as slang, fillers, swear words, and netspeak. Informality was measured as the proportion of informal words (0 to 100 continuous values) based on the informality dictionary provided by LIWC
 - Increase in **politeness** in political discussions

- Politeness is operationalized as the likelihood that the tweet is polite (0 to 100 continuous values) based on Stanford's politeness API.
- Deliberative quality:
 - Justification (whether the author offers a justification, either based on personal experiences, values, and feelings or on data, links, and facts):
 - This study developed a supervised learning classifier to label a tweet as justified or not (1/0)
 - Constructiveness (authors' attempt to move the conversation forward, build and bring consensus, and resolve conflicts)
 - A self-developed supervised learning classifier to label a tweet as constructive or not (1/0)
- Interactivity:
 - Reciprocity (intent to elicit a response or further information)
 - A self-developed supervised learning classifier to label a tweet as constructive or not (1/0)
 - Empathy and respect (acknowledgement of or sensitivity to others)
 - A self-developed supervised learning classifier to label a tweet as containing empathy or respect or not(1/0)

d. Analysis

- Interrupted time series regression model
 - $Y_{\text{feature},t} = \beta_0 + \beta_1 T + \beta_2 X_t + \beta_3 T X_t$ (T = time distance from 7 Nov, 2017; X, dummy variable, 0 = tweet published before, 1 = tweet published after)
 - Bootstrapped the analysis for 100 iterations, to tackle the inconsistency of the numbers of tweets on each day and avoid Type 1 errors

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- Manipulation check:
 - The average number of words per tweet increased significantly after the character limit intervention
- Observations from the curve plots:
 - No discontinuities at the intervention were observed for offensiveness. (offensiveness did not change significantly from pre- to post- intervention)
 - Incivility, swear words, and informal words showed a **discontinuous decrease** on the day of intervention.
 - Politeness, justification, constructiveness, and reciprocity showed a discontinuous increase
 - Empathy and respect decreased discontinuously.
- RQ1:
 - Generally, discussions become more civil after intervention.
 - Two out of four measures (uncivil words and swear words) of uncivil behavior significantly changed post-intervention across all specifications.
- H1:

- Supported: with the increase in the character limit per post, political discussions on Twitter became more civil.
- H2a:
 - Supported: compliers were more likely to use justifications after the character limit change across all model specifications. They were also more likely to write constructive tweets across all the specifications
- H2b:
 - Partially supported: Across all model specifications, compliers were more likely to manifest reciprocity in their political discussions after the character limit change. However, less likely to show empathy and respect.

b. Include Key Figures/Tables

Table 2 The Deliberative Criteria Used to Measure the Quality of Political Discussion

Dimension	Operationalization	Source
Uncivil behavior		
Incivility	Whether a tweet contains abuses and vulgarity, threats, or exaggeration (1/0 binary coding)	This study
Offensiveness	Whether a tweet contains abuses and racial slurs (1/0 coding)	Davidson et al. (2017)
Uncivil words	The proportion of a tweet which comprises words from a curated list of offensive words (0 to 100 continuous values)	Muddiman, McGregor, & Stroud (2018)
Swear words	The proportion of a tweet which comprises words from a curated list of swear words (0 to 100 continuous values)	Linguistic inquiry and word count (Pennebaker, Booth, et al., 2015)
Deliberative behavior		
Politeness	Likelihood that the tweet is polite (0 to 100 continuous values)	Stanford's politeness API (Danesco-Niculescu-Mizil et al., 2013)
Informal words	The proportionate use of informal language, such as slang, fillers, swear words, and netspeak (0 to 100 continuous values)	Linguistic inquiry and word count (Pennebaker, Booth, et al., 2015)
Justification	Whether the tweet offers evidence in the form of feelings, experiences, or facts (1/0 binary coding)	This Study
Constructiveness	Whether the tweet offers a fact-check, search for common ground, or a solution (1/0 binary coding)	This Study
Reciprocity	Whether the tweet is intended to elicit an answer, information, or feedback (1/0 binary coding)	This Study
Empathy & respect	Whether the tweet contains empathy or respect for others (1/0 binary coding)	This Study

Note: API = Application Programming Interface.

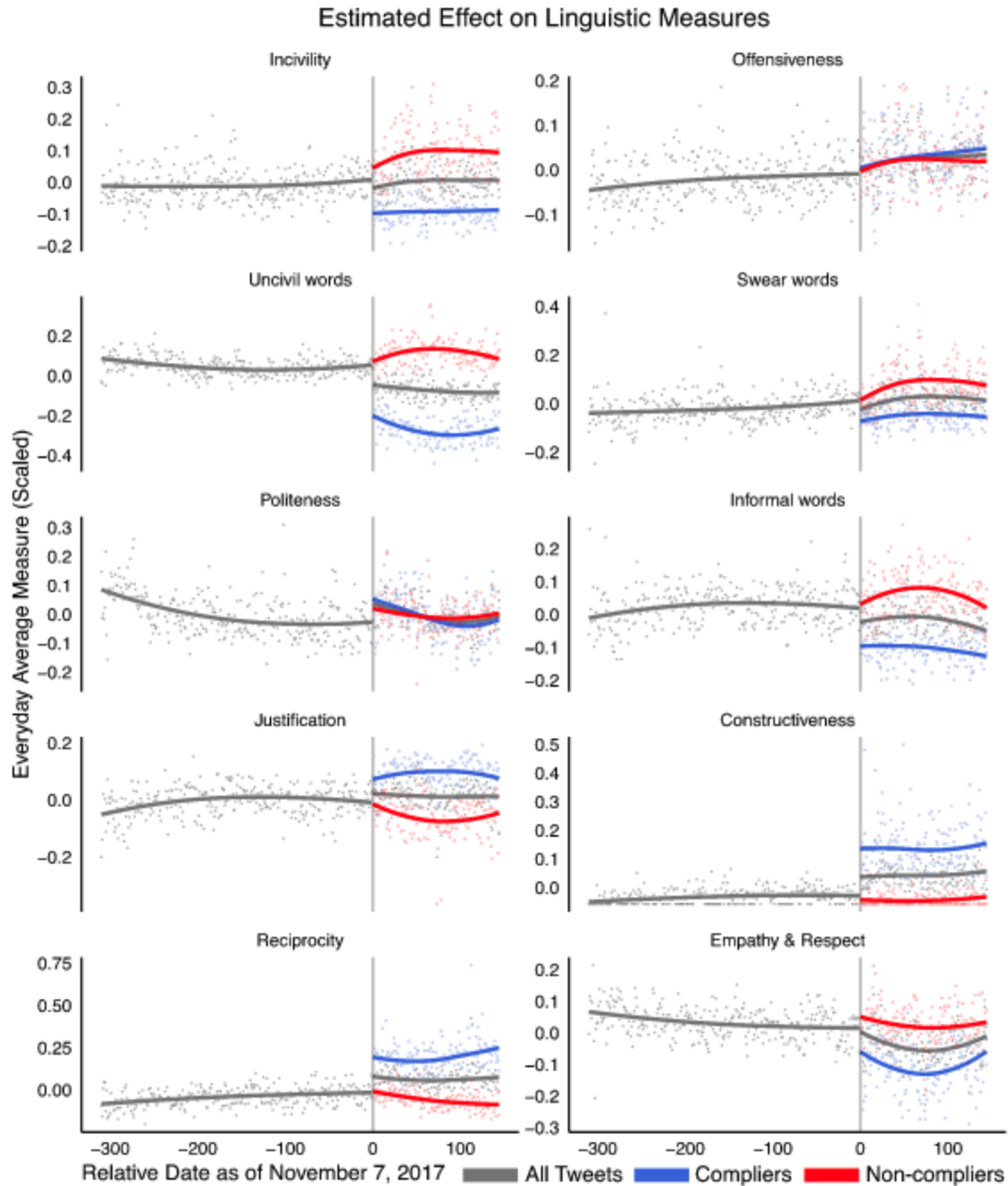


Figure 1 Daily trends of uncivil and deliberative behavior from January 2017 to March 2018 for political discussions on Twitter. The solid curve is generated by locally weighted (LOESS, locally estimated scatterplot smoothing) regression of the uncivil and deliberative attributes on sequential day numbers, with no adjustment for covariates. Data were aggregated as day-level means here to facilitate visualization.

c. Authors' Conclusions

- While Twitter's implementation of a character limit change to double the length of tweets led to less uncivil political discussions and more deliberative political discussions, it also decreased empathy and respect among the discussants.

4. Discussion

a. Implications

- The potential impact of affordances on online political communication in general.
 - Affordances can potentially trigger a change in media effects and the perceived credibility of online messages
- A change in affordances can trigger changes in online social relationships (empathy and respectfulness)
- The finding could raise concerns that a change in affordances creates new social norms and structural inequalities in the online public sphere.
 - Maybe these changes only matter to politically engaged users or elite users (implied the results comparing compliers and noncompliers)
- Online political discussions are not as toxic as imagined. Only about 4% were predicted to be uncivil

b. Limitations

- Internal validity issues of quasi-experimental design
 - Other potential confounding factors: 8 Nov 2017 was the first anniversary of the 2016 US presidential election.

c. Future Directions

- Conduct more time series analyses with social media data to better understand how social norms on social media evolve
 - Analyses can go beyond individual tweets but extend to conversation-level, including tweets and replies in one thread
 - Recommend to use Twitter's firehose stream
- Further examine how these change the overall political engagement
- Extend the research to cross- or multi-lingual contexts

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Response Letter (~600 words)

- Your assessment of the study
- Weaknesses
- Other notes

This study applied a time series design to conduct automated content analysis, exploring how political discussions on Twitter evolved after Twitter changed its message length limit from 140 words to 280 words.

I assess this study as methodologically robust and theoretically significant. The researchers combined computational methods and affordance theory. Since my previous understanding of this theory comes from a critical or cultural perspective, I find the application of quantitative and computational approaches to extend and develop affordance theory quite eye-opening and impressive.

My favorite section of this study is the method section. The paper has a robust section that documents operationalization, measurements and robustness tests in detail, from which I learned a great deal. First, the operationalization of different dependent variables is designed and documented in a detailed and literature-based way. I am also impressed by the inclusion of covariates and the robustness and placebo tests. The researchers included tests of compliers/non compliers, within-subject effects to compare subjects with themselves, and a placebo test in an unrelated time period in 2016. (I am still confused about what a bandwidth test is.) These tests help mitigate confounding effects and bring the conclusions closer to causal inference, since this study is based on observational data rather than experimental treatment. Although this study is designed as a quasi-experiment, and there might be confounding factors, such as 8 Nov 2017 being the first anniversary of the 2016 US presidential election, as noted by the authors, This is not a particularly noteworthy weakness due to the careful design.

When reading this paper, I still have some questions. First, I noticed some overlap between incivility and politeness/formality. The measurements of both contain swear words. Why is the effects on incivility framed as an explorative research question while a hypothesis is proposed for politeness/formality? Is this only due to conflicting findings on incivility in previous research? Why did the researchers compare four models of incivility? I also noticed that some measures range from 0 to 100 as continuous variables, while some are coded as binary variables. Is this because different analytical approaches, such as supervised machine learning or dictionary-based text analysis were applied? Does this difference matter for the analysis?

The findings of this study are very interesting, meaningful and significant. It emphasizes social media design features and affordances and their possibly underestimated impact on political

discussion and democratic discourses. Because this study is situated in the 2017 context, after the 2016 political election, when the public began to recognize the enormous manipulative power of giant tech companies, these findings were particularly groundbreaking. However, I personally feel that it would have been better if the authors had provided more interpretation and possible explanations for why increasing the character limit had such impacts. Given the several conflicting findings outlined in the literature review part, it would be really interesting to explore why this study reveals a distinct pattern. Is it due to platform differences? I like how this paper compares compliers and non-compliers and highlights in the discussion that affordance changes may primarily affect elite users. I like the discussion of the concern that the affordance changes may not be sufficient enough to reduce the structural inequalities. At the same time, this implies limitations of techno-determinism, and stresses the importance of human agency.

Personally, I really like this research. I first read it in 2022 when I was writing my master's SOP and it gave me a great deal of inspiration. I have recently noticed several studies using time series analyses to explore the impacts of internet laws, policy or changes of social media design on public discourse. For example, some papers analyze how Chinese social media's display of users' IP addresses influence the politeness and civility of posts. I also recently found a computational study examining how enactment of age verification laws changes people's digital behaviors including VPN usage and visits to porn sites.